Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Food Science | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Lactose is made of which two sugars?

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. Glucose and fructose

B. Two glucose units

C. Galactose and fructose

D. Glucose and galactose

Correct: D

2. (2 points) Casein clumps most at about which pH?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. 4.6

B. 6.7

C. 7.4

D. 9.0

Correct: A

3. (2 points) HTST pasteurization uses:

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. 140 °C for 30 minutes

B. 72 °C for 15 seconds

C. 63 °C for 15 seconds

D. 100 °C for 1 hour

Correct: B

4. (2 points) Which product is safe for someone with lactose intolerance but NOT for someone with milk allergy?

Unit: Lactose intolerance, allergies, alternatives, and the notebook.
Worked-case data: A student plans to test how vinegar volume affects curd mass. Design one notebook experiment.
Separate investigation observations:
Person 1: Bloating after milk; fine with yogurt and lactose-free milk.
Person 2: Hives within minutes of a tiny amount of cheese.
Test: Person 2 has antibodies to casein.

A. Water

B. Rice milk

C. Lactose-free cow’s milk

D. Oat milk

Correct: C

5. (2 points) Forming one triglyceride releases how many water molecules?

Unit: Fats: saturation, esters, and emulsions.
Worked-case data: A team churns 150 mL of cream (mass 148 g) and collects 52 g of butter. If butter is about 80% fat, estimate the cream's fat percentage.
Separate investigation observations:
Trial 1: Cream at 4 °C: no butter after 15 minutes of shaking.
Trial 2: Cream at 13 °C: butter formed in 8 minutes.
Trial 3: Cream at 25 °C: soft, greasy butter; cloudy buttermilk.

A. None

B. Three

C. One

D. Two

Correct: B

6. (2 points) Which solute lowers freezing point more per mole?

Unit: Freezing point depression and ice cream.
Worked-case data: Estimate the freezing point of meltwater in which 2.0 mol of NaCl dissolves per kilogram of water.
Separate investigation observations:
Trial 1: Milk, 30 mL salt: soft slush, 2 cm tall.
Trial 2: Milk, 90 mL salt: firmer, 4 cm tall.
Trial 3: Cream, 90 mL salt: firm and smooth, 7 cm tall.

A. They are equal

B. Neither

C. NaCl

D. Sugar (sucrose)

Correct: C

7. (2 points) On a label, 25% DV of a nutrient is considered:

Unit: Nutrition and food labels.
Worked-case data: A label lists 300 mg of calcium per serving and a Daily Value of 1,300 mg. A student drinks 3 servings. What percent DV is that, and is one serving “high”?
Separate investigation observations:
Milk A: 8 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk B: 0 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk C: 5 g fat, 24 g total sugars, 12 g added sugars, 25% DV calcium.

A. Low

B. Moderate only

C. Not meaningful

D. High

Correct: D

8. (2 points) A triglyceride contains glycerol and:

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. Two glucose units

B. One phosphate and one sugar

C. Three fatty acids

D. Three amino acids

Correct: C

9. (2 points) Why can boiling fresh milk not cause curdling by itself?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. Whey proteins cannot denature

B. Boiling raises pH to 10

C. Lactose prevents clumping

D. Casein is heat-stable at normal pH

Correct: D

10. (2 points) Why does yogurt thicken?

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. Lactic acid lowers pH and casein gels

B. Fat crystallizes

C. Sugar is added

D. Water evaporates

Correct: A

11. (2 points) A dairy label gives 9 g carbohydrate, 4 g protein and 3 g fat per serving. Using 4/4/9 kcal per gram, calculate energy.

answer: 79 kcal (2 points).

Numerical target 79; absolute tolerance 0.158. Accepted units: , kcal.

12. (2 points) A milk experiment begins with 110 g milk and produces 11 g drained curd. Find curd mass as percent of starting milk mass.

answer: 10 % (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , %.

13. (2 points) A cream experiment starts with 110 g cream and produces 22 g butter. Find butter mass yield percentage.

answer: 20 % (2 points).

Numerical target 20; absolute tolerance 0.04. Accepted units: , %.

14. (2 points) A team obtains 3 g curd while the supervisor obtains twice that mass. Using mass ratio times 10, calculate the task score.

answer: 5 (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , .

15. (2 points) A synthetic ice-cream mound stands 90 mm high for the specified five seconds. Using 0.1 points/mm, calculate its height score.

answer: 9 (2 points).

Numerical target 9; absolute tolerance 0.018. Accepted units: , .

16. (2 points) Explain why adding salt to the surrounding ice bath helps cool a sealed dairy mixture.

answer: Dissolved salt lowers the freezing point of the external water/ice mixture, allowing a colder bath and heat transfer out of the sealed dairy mixture. (2 points).

17. (2 points) Explain how acid addition can cause casein-rich curds to separate from milk.

answer: Acid lowers pH toward conditions that reduce casein-micelle stability, promoting aggregation; the liquid whey can then be drained from the curds. (2 points).

18. (2 points) Distinguish lactose intolerance from a milk-protein allergy.

answer: Lactose intolerance concerns inadequate digestion of lactose by lactase; milk-protein allergy involves an immune response to milk proteins. They are different mechanisms. (2 points).

19. (2 points) Does lactase treatment remove all carbohydrate from milk? Explain.

answer: No. Lactase hydrolyzes lactose into smaller sugars; it does not simply remove the sugar mass. (2 points).

20. (2 points) Why must a dairy-curd comparison standardize draining time and starting milk quantity?

answer: Retained liquid and different initial sample amounts can change measured curd mass independently of the experimental variable. (2 points).

21. (2 points) Use Figure F1. Estimate serving energy with 4/4/9 kcal per gram.

One serving of a teaching dairy product

Item	Per serving
Total fat	4 g
Protein	8 g
Total carbohydrate	13 g
Total sugars	10 g
Includes added sugars	3 g
Calcium	260 mg; DV1300 mg

F1. Dairy nutrition label (original synthetic teaching figure)

answer: 120 kcal (2 points).

Numerical target 120; absolute tolerance 0.36. Accepted units: , kcal.

22. (2 points) Use Figure F1. Calculate calcium percent Daily Value.

Refer to F1. Dairy nutrition label (original synthetic teaching figure) shown with the first question in this figure group.

answer: 20 % (2 points).

Numerical target 20; absolute tolerance 0.06. Accepted units: , %.

23. (2 points) Use Figure F1. How many grams of sugars are not labeled as added?

Refer to F1. Dairy nutrition label (original synthetic teaching figure) shown with the first question in this figure group.

answer: 7 g (2 points).

Numerical target 7; absolute tolerance 0.021. Accepted units: , g.

24. (2 points) Use Figure F2. Compute percent mass yield in A.

All trials use100 g starting milk and5 min draining

Trial	Milk type	Vinegar (mL)	Drained curd (g)
A	Whole	2	13
B	Skim	2	10
C	Whole	4	16

F2. Dairy curd yield experiment (original synthetic teaching figure)

answer: 13 % (2 points).

Numerical target 13; absolute tolerance 0.039. Accepted units: , %.

25. (2 points) Use Figure F2. Which displayed pair best isolates vinegar volume with milk type matched?

Refer to F2. Dairy curd yield experiment (original synthetic teaching figure) shown with the first question in this figure group.

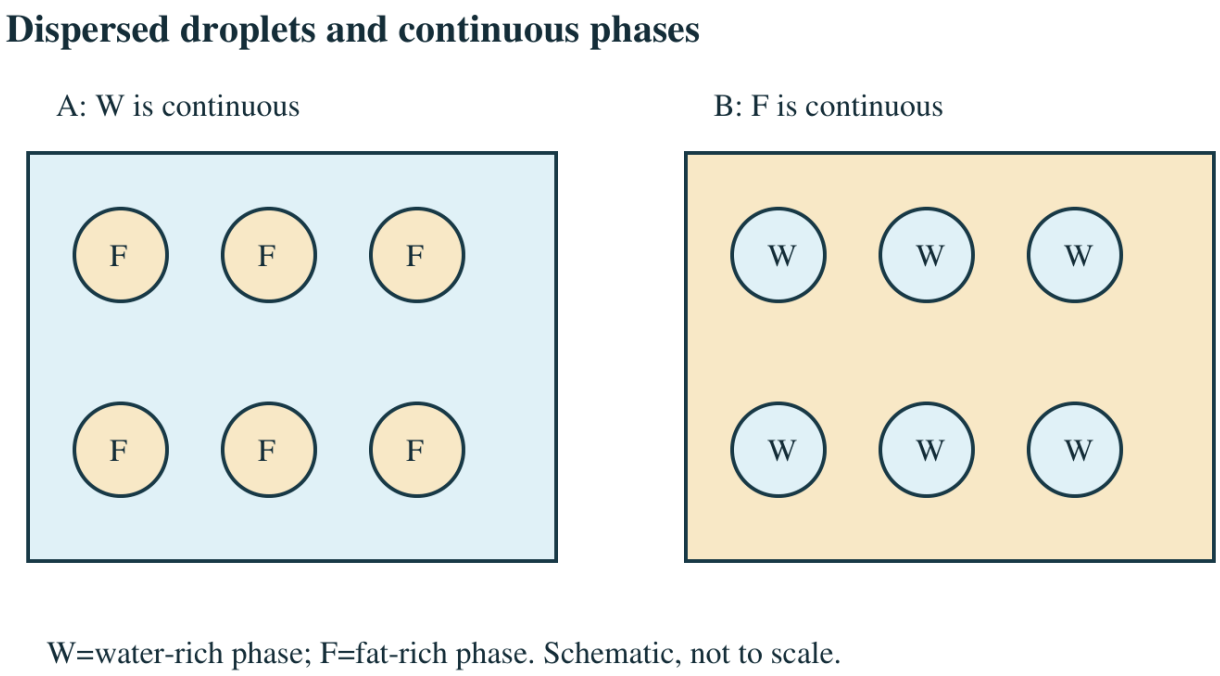
answer: A and C; both use whole milk and differ in vinegar amount while starting mass and draining time are matched. (2 points).

26. (2 points) Use Figure F2. Why would failing to standardize draining time weaken this comparison?

Refer to F2. Dairy curd yield experiment (original synthetic teaching figure) shown with the first question in this figure group.

answer: Different retained whey can change measured curd mass independently of curd solids or the intended experimental variable. (2 points).

27. (2 points) Use Figure F3. Which panel represents the usual oil-in-water organization of milk?



F3. Dairy emulsion microstructures (original synthetic teaching figure)

answer: A. (2 points).

28. (2 points) Use Figure F3. Which panel represents the water-in-fat organization associated with butter?

Refer to F3. Dairy emulsion microstructures (original synthetic teaching figure) shown with the first question in this figure group.

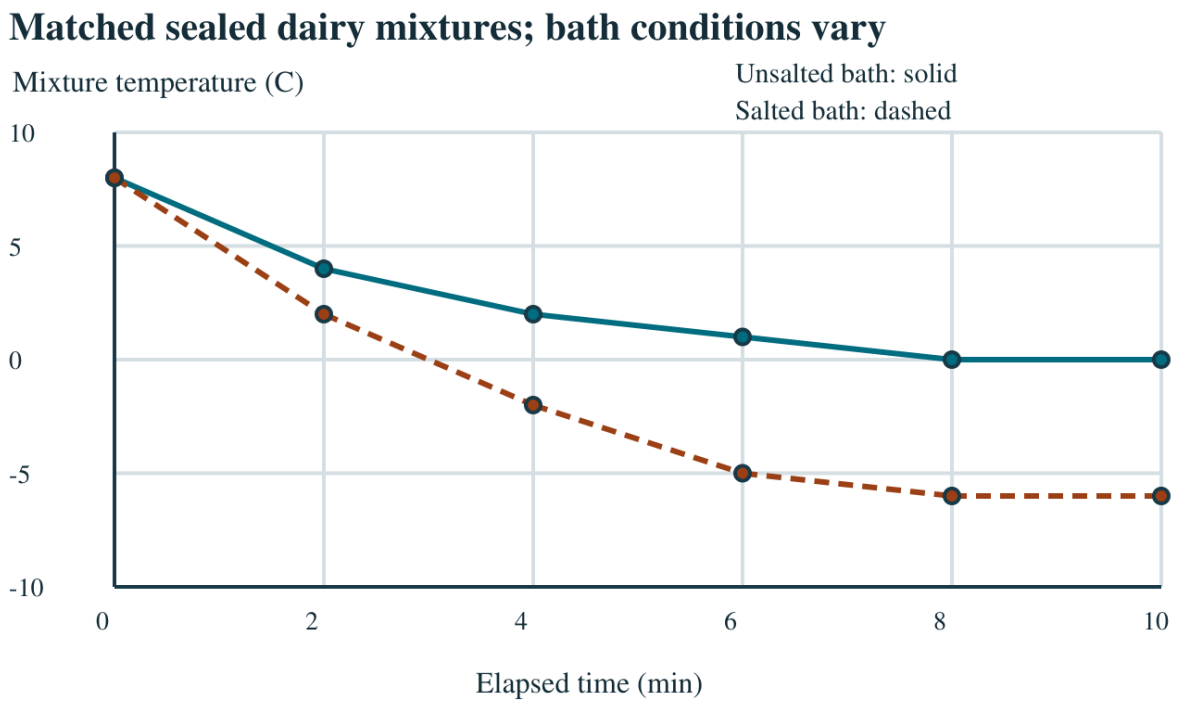
answer: B. (2 points).

29. (2 points) Use Figure F3. Describe the phase-organization change represented by A to B during churning.

Refer to F3. Dairy emulsion microstructures (original synthetic teaching figure) shown with the first question in this figure group.

answer: Fat becomes the continuous phase and water-rich material becomes dispersed droplets; this is a phase inversion rather than removal of all water. (2 points).

30. (2 points) Use Figure F4. Read the salted-bath mixture temperature at6 min.



F4. Ice-bath cooling curves (original synthetic teaching figure)

answer: -5 C (2 points).

Numerical target -5; absolute tolerance 0.015. Accepted units: , C.

31. (2 points) Use Figure F4. Find the temperature difference unsalted minus salted at8 min.

Refer to F4. Ice-bath cooling curves (original synthetic teaching figure) shown with the first question in this figure group.

answer: 6 C (2 points).

Numerical target 6; absolute tolerance 0.018. Accepted units: , C.

32. (2 points) Use Figure F4. Explain the physical role of salt in the surrounding bath.

Refer to F4. Ice-bath cooling curves (original synthetic teaching figure) shown with the first question in this figure group.

answer: Dissolved salt lowers the freezing point of bath water, allowing an ice-water bath below0C and heat transfer out of the sealed dairy mixture; it need not enter the food. (2 points).